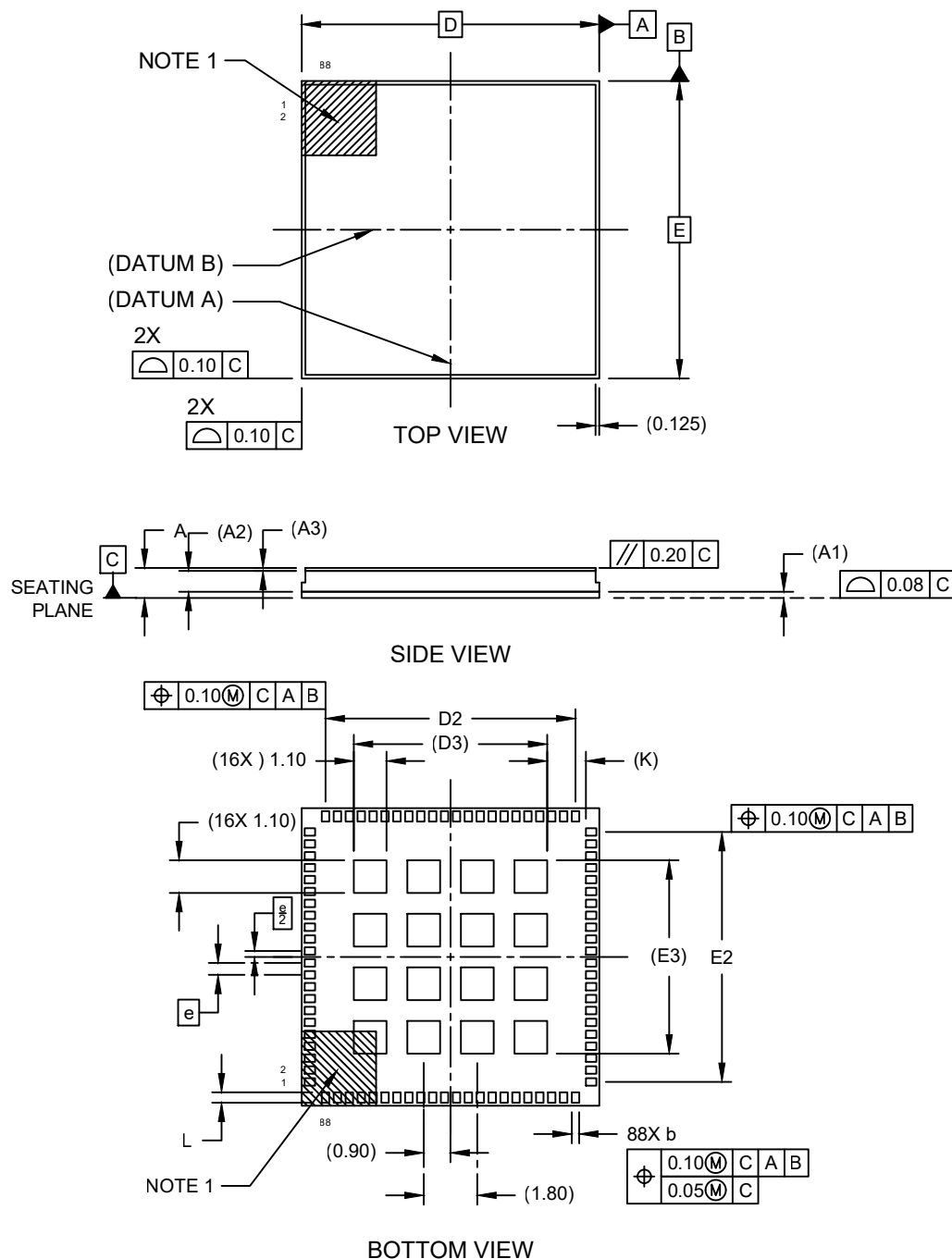


Packaging Diagrams and Parameters

88-Lead Thin Fine-Pitch Land Grid Array (4NW) - 10x10x1.1 mm Body [TFLGA] With Heat Sink and Multiple Exposed Pads

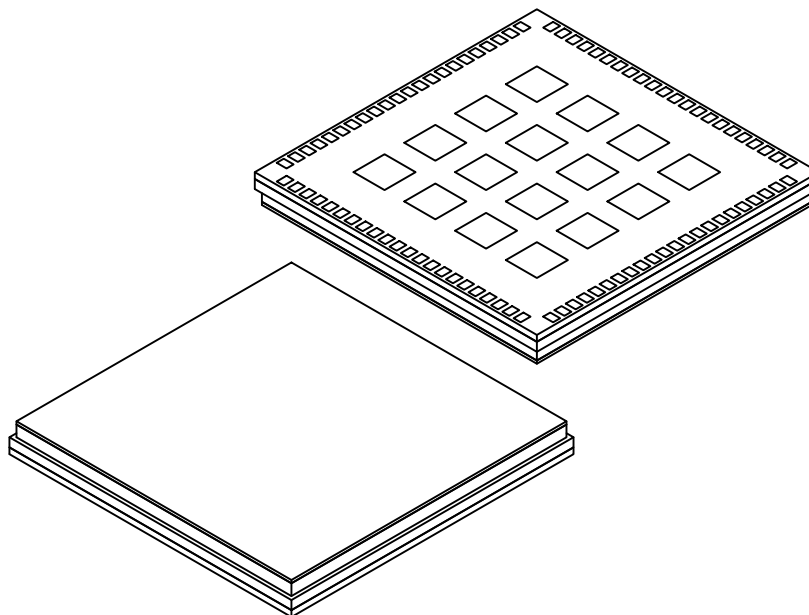
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

88-Lead Thin Fine-Pitch Land Grid Array (4NW) - 10x10x1.1 mm Body [TFLGA] With Heat Sink and Multiple Exposed Pads

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	88		
Pitch	e	0.40 BSC		
Overall Height	A	–	–	1.10
Substrate	A1	0.21 REF		
Mold Thickness	A2	0.70 REF		
Heat Sink Thickness	A3	0.10 REF		
Overall Length	D	10.00 BSC		
Lead Edge Center to Center	D2	8.40 BSC		
Exposed Pad Array Length	D3	6.50 REF		
Overall Width	E	10.00 BSC		
Lead Edge Center to Center	E2	8.40 BSC		
Exposed Pad Array Width	E3	6.50 REF		
Terminal Width	b	0.20	0.25	0.30
Terminal Length	L	0.30	0.35	0.40
Terminal-to-Exposed-Pad	K	1.30 REF		

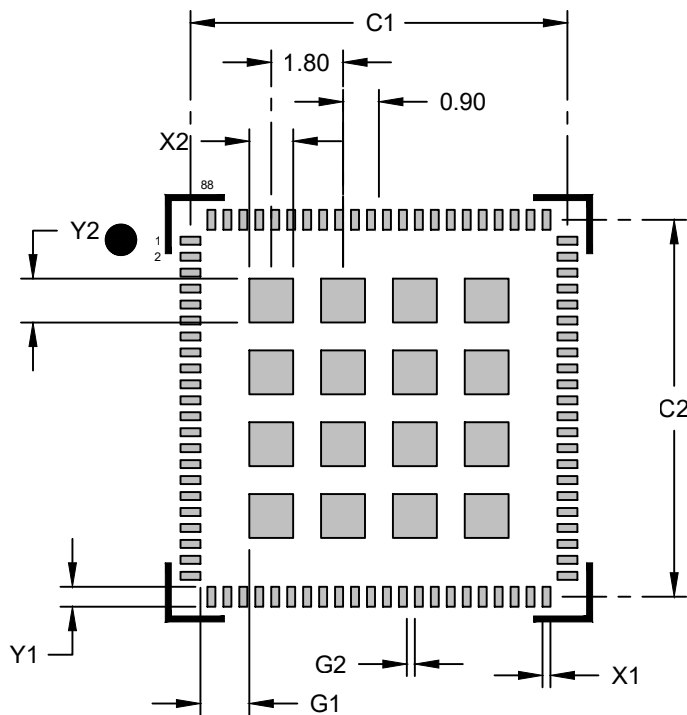
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

88-Lead Thin Fine-Pitch Land Grid Array (4NW) - 10x10x1.1 mm Body [TFLGA] With Heat Sink and Multiple Exposed Pads

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Pad Width (X16)	X2			1.10
Center Pad Length (X16)	Y2			1.10
Contact Pad Spacing	C1		9.45	
Contact Pad Spacing	C2		9.45	
Contact Pad Width (X88)	X1			0.20
Contact Pad Length (X88)	Y1			0.50
Contact Pad to Center Pads (X88)	G1	1.23		
Contact Pad to Contact Pad (X84)	G2	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-02593 Rev B